

Greeks Ratios and LEAPS

Options Strategy Discussion — Vega, Gamma, Delta, and LEAP Structures

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1. Vega Traded / Vega Outstanding — Detecting Mispricings

1.1 Definition

This ratio compares **flow** (how much vega is changing hands today) to **stock** (how much vega is sitting in open interest). It is essentially a turnover rate for volatility exposure on a given name.

- **Vega Traded** = sum of vega across all contracts traded in a session
- **Vega Outstanding** = sum of vega across all open interest

The ratio measures how actively the market is re-pricing or repositioning its volatility exposure.

1.2 Interpretation

High ratio (e.g., >0.3–0.5):

- Someone is aggressively trading vol relative to what's outstanding
- Signals new information being priced in — earnings, M&A, FDA events, macro catalysts
- Existing open interest is being “turned over”
- **Opportunity:** If the ratio spikes but IV hasn't moved yet, the market may be slow to reprice

Low ratio (e.g., <0.05):

- Stale positioning — nobody is actively re-pricing vol
- IV may be anchored to outdated assumptions
- **Opportunity:** If you have a fundamental view that vol should be different, you may get favorable entry prices

1.3 Practical Edge Cases for Detecting Mispricings

1. **Ratio spikes + IV flat** — Flow is happening but the surface hasn't moved. Someone knows something or is hedging quietly. Investigate direction of flow.
2. **Ratio spikes + skew distortion** — Heavy activity in specific strikes/tenors shifts supply-demand locally. Look for dislocations for relative value plays (calendar spreads, butterfly).
3. **Low ratio + upcoming catalyst** — If events are approaching but nobody is trading vol, IV may be underpriced. The OI was built for a different regime.
4. **Cross-name comparison** — Compare the ratio across correlated names (e.g., GLD vs SLV). If one shows heavy vega turnover and the other doesn't, the laggard may need to catch up.
5. **Term structure signal** — Calculate the ratio per expiry. If front-month vega turnover is 10x the back-month, the market is pricing a near-term event.

2. Concrete Examples: SPY vs AA Jan'27 LEAP

2.1 SPY — The Liquid Beast

SPY at ~\$600, normal trading day, March '26 expiry (~30 DTE):

Strike	OI	Vega/contract	Vega Outstanding	Volume	Vega Traded
590 P	120,000	\$0.18	\$21,600	35,000	\$6,300
595 P	85,000	\$0.22	\$18,700	12,000	\$2,640
600 C	200,000	\$0.25	\$50,000	80,000	\$20,000
605 C	95,000	\$0.21	\$19,950	15,000	\$3,150
610 C	60,000	\$0.15	\$9,000	8,000	\$1,200

Normal day ratio: ~0.28 (25–30% turnover baseline for SPY).

FOMC day: Volume triples, ratio jumps to ~0.80. If IV only ticks up 1% despite this, the surface is lagging the flow — potential edge buying vol.

Key for SPY: baseline turnover is already high, so you need a **relative spike** vs the 20-day average.

2.2 AA (Alcoa) — Jan'27 LEAP

AA at ~\$38, Jan'27 expiry, very different dynamics:

Strike	OI	Vega/contract	Vega Outstanding	Volume	Vega Traded
30 C	4,200	\$0.08	\$336	15	\$1.20
35 C	8,500	\$0.11	\$935	40	\$4.40
40 C	6,800	\$0.12	\$816	120	\$14.40
45 C	3,200	\$0.09	\$288	25	\$2.25
35 P	2,100	\$0.10	\$210	10	\$1.00

Normal day ratio: 0.009 — essentially dead.

After aluminum price spike: Ratio jumps to 0.17 (a 19x increase). The 50 strike traded 2x its entire OI in one day — conviction flow.

If Jan'27 IV stays flat despite this flow, the surface is lagging — mispricing opportunity.

2.3 Side-by-Side Comparison

Metric	SPY Mar '26	AA Jan '27
Baseline ratio	0.25–0.30	0.005–0.01
Signal threshold	>0.70	>0.10
IV repricing speed	Minutes	Hours to days
Edge duration	Very short	Can persist 1–3 days
Best use	Event scalping, gamma	Structural vol mispricing
Risk	Fast; edge disappears quickly	Wide bid-ask, illiquidity

3. Optionality = Vega + Gamma

An option's total "optionality value" has two engines:

- **Vega channel** — IV moves, option reprices, underlying does nothing
- **Gamma channel** — underlying moves, delta changes non-linearly, option captures convexity even if IV stays flat

3.1 Concrete Example on AA

Buy the Jan'27 40C at \$4.50 with AA at \$38:

Scenario A — IV repricing only (vega leg): AA sits at \$38, IV moves 42% to 48%. Gain ~\$0.72 from vega alone.

Scenario B — Breakout only (gamma leg): AA breaks to \$44, IV flat. Option goes from \$4.50 to ~\$8.20. The \$3.70 gain exceeds delta prediction (\$2.52), with \$1.18 coming from gamma P&L.

Scenario C — Both fire: \$3.70 from gamma/delta + \$0.72 from vega = \$4.42 on a \$4.50 entry.

3.2 Gamma Traded / Vega Traded — Two Lenses

For a swing/breakout trader who also sells vol:

Gamma ratio	Vega ratio	Interpretation
High	Low	Directional bet, vol not repricing — buy options (cheap vega)
Low	High	Pure vol trade — sell vol via spreads if repricing overshoots
High	High	Full repositioning — be careful
Low	Low	Nothing happening — wait or sell premium

Breakout candidates: Gamma ratio spikes, vega ratio stays low — buy with cheap IV.

Vol selling candidates: Vega ratio spiked recently but falling back, IV still elevated — sell credit spreads into mean reversion.

4. Sell LEAP Put + Short-Term Protection on AA

4.1 The Structure

- **Sell Jan'27 35P** — collect ~\$4.50 premium
- **Buy Mar'26 33P** — pay ~\$0.80 premium
- Net credit: \$3.70

By put-call parity, the short LEAP put is synthetically: long AA at 35 + short Jan'27 35C. That's a covered call without touching the stock — using margin instead of ~\$3,800 in capital.

4.2 Greek Profile

Greek	LEAP Short Put	Protection Put	Net Position
Delta	~-0.35	~+0.20	~-0.15 (slightly bullish)
Gamma	~0	+0.04	+0.04 (near-term convexity)
Theta	+\$0.02/day	-\$0.03/day	~-\$0.01 (roughly flat)
Vega	-\$0.12	-\$0.03	-\$0.12 (short vol)

Key: gamma and vega are on **different expirations** — decoupled by design.

4.3 Rolling the Protection

AA does...	Action	Result
Breaks out higher	Let protection expire, buy next month cheaper	Net credit improves
Chops sideways	Roll to next month at similar cost	LEAP theta works for you
Drops hard	Protection goes ITM, offsets LEAP loss	Hedged; reassess

4.4 Key Risks

1. **Margin expansion on gap down** — Broker recalculates margin; compounding across 10–15 positions
2. **Protection gap** — Between expiry and roll, you're briefly naked
3. **Vega expansion on the LEAP** — If Jan'27 IV goes from 42% to 55%, LEAP put loses ~\$1.56 from vega alone

5. Buy LEAP Call vs Sell LEAP Put — Decision Framework

5.1 Structural Differences

	Buy Jan'27 Call	Sell Jan'27 Put
Capital	Debit (you pay)	Credit (you collect)
Vega	Long — want IV to rise	Short — want IV flat/down
Gamma	Long (small, positive)	Short (small, negative)
Theta	Against you	For you
Max loss	Premium paid (defined)	Strike minus premium (large)
Win condition	Big move up	Not moving down much

5.2 Decision Criteria

Sell the LEAP Put when:

1. Vega ratio is low (<10%) — nobody repricing vol
2. IV is at or above fair value (IV percentile >40th)
3. No sharp breakout expected
4. Income generation goal

Buy the LEAP Call when:

1. Vega ratio spiking (>15–20%) with IV still flat
2. IV is cheap (IV percentile <20th)
3. Breakout setup forming
4. Catalyst ahead not priced in

5.3 Decision Flowchart

Is IV percentile > 40th AND Vega ratio < 10%?

YES -> environment favors SELLING vol

Is the stock ranging / no breakout setup?

YES -> SELL LEAP PUT (+ protection)

NO -> Sell put anyway (income compromise)

NO -> environment favors BUYING vol

Is there a breakout setup or catalyst?

YES -> BUY LEAP CALL

NO -> WAIT, neither trade has edge

5.4 AA Current Market Data (Feb 2026)

Data point	Value	Implication
Vega ratio	5%	Dead — nobody repricing vol
Volume vs 30d avg	-16%	Quieter than usual
OI P/C	0.83	Mild bullish positioning
IV change (30d)	+1%	Noise

Data point	Value	Implication
IVP (52w)	95%	IV higher than 95% of past year — expensive
HVP (52w)	87%	Stock genuinely moving — vol justified
IVP – HVP gap	+8%	IV slightly rich to realized, not extreme

Conclusion: Sell the LEAP put. IV at 95th percentile = fat premium. Go further OTM than usual (20-delta instead of 30-delta) since elevated IV inflates premiums across all strikes. Reduce protection cost via put spreads.

6. LEAP Put Delta Below 50 — Forward Price and Lognormal Effects

6.1 The Observation

AA last price: \$60.51. The Jan'27 60 Put has delta = 38, while the Feb'26 60 Put has delta = 48. Why is the LEAP put delta so far below 50 for an apparently ATM option?

6.2 Options Price Off the Forward, Not Spot

$$F = S \times e^{(r-q) \times T}$$

With $S = 60.51$, $r = 4.3\%$, $q = 0.67\%$ (AA dividend yield per IBKR):

Feb '26 ($T = 0.08$): $F = 60.51 \times e^{0.032 \times 0.08} = \60.67

The 60 strike is barely OTM vs forward. Delta ≈ 48 .

Jan '27 ($T = 0.92$): $F = 60.51 \times e^{0.032 \times 0.92} = \62.56

The 60 strike is $\sim \$2.56$ OTM vs forward. This explains part of the delta drop.

6.3 The Lognormal Effect — Delta Is Not Probability of ITM

This is the critical insight. Black-Scholes put delta uses $N(-d_1)$. Probability of expiring ITM uses $N(-d_2)$. They differ by:

$$d_1 - d_2 = \sigma\sqrt{T}$$

For AA Jan'27: $0.45 \times \sqrt{0.92} = 0.43$. A massive wedge.

Full Computation

$$d_1 = \frac{\ln(60.51/60) + (0.043 - 0.0067 + 0.45^2/2) \times 0.92}{0.45 \times \sqrt{0.92}} = \frac{0.0085 + 0.1265}{0.4316} = 0.313$$

$$d_2 = d_1 - \sigma\sqrt{T} = 0.313 - 0.432 = -0.119$$

$$\text{Put delta} = -N(-d_1) = -N(-0.313) = -0.377 \approx 38\%$$

$$P(\text{ITM}) = N(-d_2) = N(0.119) = 0.547 \approx 55\%$$

The 60 put has a 55% chance of expiring ITM but a delta of only 38.

6.4 Why the Divergence?

Delta measures: “if the stock moves \$1 right now, how much does the option price change?”

P(ITM) measures: “what fraction of possible outcomes end below the strike?”

These diverge because of **asymmetry in payoff magnitude** under the lognormal distribution.

The 60 put expires ITM 55% of the time. But most of those ITM outcomes are **shallow** — stock at \$59, \$58, \$57. The deep ITM outcomes (stock at \$40, \$30) are rare because the lognormal left tail is compressed (prices can’t go below zero).

Meanwhile, the 45% of paths above \$60 include outcomes at \$80, \$100+ where the put is worthless.

When the stock moves from \$60.51 to \$61.51 today:

- The high-payoff ITM scenarios (stock at \$30) are barely affected by \$1 spot move
- The near-strike scenarios (\$59 vs \$60) are affected, but their payoff is small
- **Delta weights by payoff magnitude. P(ITM) just counts paths.**

For short duration (Feb '26), $\sigma\sqrt{T} = 0.127$ — the d1/d2 gap is small, so delta $\approx$ P(ITM). For LEAPs with high IV, the gap is enormous.

6.5 Practical Implication for LEAP Put Sellers

What the screen says	What it actually means
Delta = 38	Feels OTM, moderate risk
P(ITM) = 55%	Actually more likely than not to expire ITM

You will be assigned more often than the 38 delta suggests. But most ITM outcomes are shallow — stock at \$59, \$58 — where your loss is small relative to premium collected. The expected value works because the probability-weighted average payoff remains moderate.

This is precisely why selling LEAP puts works as an income strategy: you are “ITM” frequently but “deeply ITM” rarely.

7. ITM at Expiry vs Losing Money — Where the Edge Actually Lives

7.1 The Apparent Paradox

If $P(\text{ITM}) = 55\%$, does that mean 55% chance of a losing trade? **No.** ITM at expiry and losing money are not the same thing.

7.2 Breakeven Is Not the Strike

If you sell the Jan'27 60P for \$6.00 (plausible at 95th percentile IV):

$$\text{Breakeven} = \text{Strike} - \text{Premium} = \$60 - \$6.00 = \$54.00$$

Stock at expiry	ITM?	P&L	Outcome
\$62	No	+\$6.00	Full profit
\$58	Yes	+\$4.00	Profitable despite assignment
\$56	Yes	+\$2.00	Profitable despite assignment
\$54	Yes	\$0.00	Breakeven
\$50	Yes	-\$4.00	Loss
\$45	Yes	-\$9.00	Significant loss

The entire zone between \$54 and \$60 is **ITM but profitable**. This is where most of the 55% ITM probability mass sits — shallow ITM outcomes.

7.3 Three Probabilities, Three Meanings

Metric	Value	What it measures
Delta	38	Payoff-weighted risk; rough proxy for probability of net loss
$P(\text{ITM})$	55%	Probability of any assignment , including profitable ones
$P(\text{Loss})$	~25-30%	Probability stock falls below breakeven (\$54)

7.4 Why Fat Premium = Real Edge

At the 95th IV percentile, you collect roughly **double** the premium vs normal IV. This widens the strike-to-breakeven gap:

- Normal IV: sell 60P for \$3.50 → breakeven at \$56.50
- 95th pct IV: sell 60P for \$6.00 → breakeven at \$54.00

That extra \$2.50 of cushion means:

- More ITM scenarios that are still profitable
- P(Loss) drops from ~35% to ~25%
- You are getting paid for volatility that statistically mean-reverts

Even if assignment happens, you are profitable unless the move is severe. The edge is not in avoiding ITM — it's in **the premium buffer absorbing shallow ITM outcomes** while vol mean-reverts in your favor.

Appendix: Key Formulas

Forward Price:

$$F = S \times e^{(r-q) \times T}$$

Lognormal Median:

$$\text{Median} = F \times e^{-0.5 \times \sigma^2 \times T}$$

Black-Scholes d1 and d2:

$$d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2) \times T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Put Delta: $\Delta_P = -e^{-qT} \times N(-d_1)$

Probability ITM (Put): $P(\text{ITM}) = N(-d_2)$

Delta vs P(ITM) gap driven by: $\sigma\sqrt{T}$ — the higher the vol or the longer the duration, the larger the divergence.